MIL-M-45103 (Ord)
18 June 1958
SUPERSEDING
FA(FC)-PD-21
13 August 1957

MILITARY SPECIFICATION

MOUNT, TELESCOPE, T219

1. SCOPE

1.1 This specification covers one type of mount, known as Telescope Mount T219. This mount incorporating manual cant correction is used with Telescope T186 for direct fire with the 90MM Self-Propelled Gun M56.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Military

MIL-F-13926	Fire-Control Material: General Specification Governing the Manufacture and Inspection of
MIL-M-13093	Mounts, Telescopes, M87A1 and T219: Packaging of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

Ordnance Corps

F8226879	Mount, Telescope, T219

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

FSC 1290

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3. REQUIREMENTS

3.1 Applicable requirements.- Applicable requirements common to all fire control instruments as specified in MIL-F-13926 shall apply.

3.2 Materials.- Materials shall be in accordance with the applicable drawings and specifications forming part of this specification.

3.3 Vibration.- With the boresight movement secured by means of the locking levers the mount shall be vibrated for a period of five minutes, plus or minus 15 seconds, at a vertical amplitude of 1/16 inch (1/8 inch total movement) and a frequency of 30 cycles per second after which the mount shall meet all requirements herein.

3.4 Adjustment.- With the mounting surfaces horizontal and vertical respectively and with the axis through the telescope spherical bearing surfaces parallel to the aforementioned mounting surfaces, the following requirements shall be met:

3.4.1 The level bubble shall be centered within the width of a vial graduation line when the telescope locating slot is vertical.

3.4.2 The cant correction adjustment shall have at least 14.5 degrees movement on each side of level position.

3.4.3 The elevation and azimuth deflection knobs shall have at least 10 mils movement on each side of the orientation of 3.4.

3.5 Locking lever.- With 60 pound inches torque applied to the locking levers, there shall be no movement in elevation or in azimuth deflection when a torque of 25 pound inches is applied to the disengaged boresight knobs.

3.5.1 Locking the locking levers with the above required torque shall result in no change of elevation or azimuth alinement.

3.6 Backlash.- With the locking levers unlocked and the boresighting knobs disengaged, backlash at the boresighting knobs shall not exceed 0.05 mil.

3.7 Operating knobs

3.7.1 After securing and then unlocking the locking levers, the running torque for elevation and azimuth deflection boresight knobs, when disengaged, shall not exceed 15 pound inches.

3.7.2 The running torque required to rotate the cant correction knob shall not exceed 4 pound inches.

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3.8 Temperature.- The mount shall be exposed to an ambient temperature of minus 40°F and plus 150°F for a period of at least five hours after which the mount shall meet the requirements of 3.7 when exposed to these temperatures. The maximum allowable torque of 3.7.1 and 3.7.2 may be increased by fifty percent.

3.9 Pilot lot.- A pilot lot representing initial production, the size of which shall be determined by the contracting officer shall be submitted to the Government inspector for acceptance inspection.

3.10 Workmanship.- Workmanship shall be of a quality representative of the highest existing production standards and shall embody best instrument practices consistent with the intended use of the item. Standards of workmanship shall be such that the mounts shall meet all requirements herein and all components and subassemblies shall be in accordance with applicable workmanship requirements specified in Specification MIL-F-13926.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions

4.1.1 Contractor inspection.- Unless otherwise specified herein, the inspections required by this specification will be performed by the contractor.

4.1.2 Contractor quality assurance system.- The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies under this contract. Results of all examinations and tests performed under the quality assurance system shall be made available to the Government. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when, as a minimum, it provides the quality assurance required by the specification and other documents referenced in this specification. The contractor will not be restricted to the inspection station or to the method of inspection listed provided that the substitute procedures will provide, as a minimum, the quality assurance required in the contractual documents. In cases of dispute as to whether certain procedures of the system provide equal assurance, the comparable procedure of this specification shall be used. The contractor shall notify the Government of, and obtain approval for, any change to the written procedure that affects the degree of assurance required by this specification or other documents referenced therein.

4.1.3 Government verification.- All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of surveillance of the operations to determine whether the practices, methods, and procedures of the written inspection plan are being properly applied, and Government product inspection to measure quality of product offered for acceptance. Deviation

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from the prescribed or agreed upon procedures, or instances of poor practices which might have an affect upon the quality of the product will immediately be called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered will be cause for suspension of acceptance until correction has been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor will designate a responsible official or officials to whom the Government inspector will report such instances.

4.2 Inspection provisions.- Applicable quality assurance provisions common to all fire control instruments as specified in specification MIL-O-13926 shall apply.

4.2.1 Lot formation.- Unless otherwise specified by the contracting officer, or his representative, inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.2.2 Examination.- The classification of defects in table I shall constitute the minimum inspection requirements to be performed prior to acceptance.

4.2.2.1 All rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrected action taken, the lot shall be resubmitted for acceptance inspection as outlined in Standard MIL-STD-105.

TABLE I - CLASSIFICATION OF DEFECTS
(Use Inspection Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105)

MAJOR	AQL 1.5	REQ. PAR.	Q.A. PAR.	TEST METHOD
1.	Level bubble	3.4.1	4.3.2.2	DV
2.	Cant correction	3.4.2	4.3.2.3	[illegible]
3.	Elevation travel	3.4.3	4.3.2.4	[illegible]
4.	Deflection travel	3.4.3	4.3.2.5	DST
5.	Backlash	3.6	4.3.2.6	DST
6.	Locking levers	3.5	4.3.2.7	Manufacture - Std. Test Equipment
7.	Operating knobs	3.7	4.3.3	Std. Test Equip.

4.2.3 Inspection equipment .- The test methods and procedures outlined in 4.3 shall be used as standards for determining the adequacy of the contractor's inspection equipment and methods utilized for inspecting the characteristics in table I. Unless otherwise specified in the contract, the contractor shall provide and maintain gages and other measuring and testing devices considered necessary for inspection of supplies for conformance to contracted requirements. The Government reserves the right to certify all such equipment and to use this equipment for Government verification inspection.

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4.3 Test methods and procedures.-

4.3.1 Vibration.- The mount containing a dummy telescope, equal in weight distribution and contact surfaces to a Telescope M91D, shall be rigidly mounted in a horizontal position using a suitable adapter and vibrated on an approved vibrating machine. This test shall meet the requirements specified in 3.3.

4.3.2 Accuracy tests.

4.3.2.1 The following test shall be conducted with the mount set up as specified in 3.4 and fastened to a suitable leveled stand. A sighting device, having a suitable reticle with bearing surfaces and locating pin of such accuracy as to adequately perform the subsequent test of this specification, shall be utilized. The line of sight of the sighting device shall be directed toward a plumbed target containing a vertical and horizontal crossline. These lines shall have a 10 mil graduation each side of their intersection. The target shall also contain two (2) radial lines, one positioned 14 degrees 30 minutes each side of the upper vertical target line. The vertex of the radial lines shall be at the intersection of the horizontal and vertical target lines.

4.3.2.2 Level bubble.- With the elevation and deflection dials of the mount set to zero position (numeral 3) on the scale indexes, and the reticle lines of the sighting device superimposed on the vertical and horizontal target lines, the level bubble shall be central within the limits specified in 3.4.1.

4.3.2.3 Cant correction.- With the mount set at zero cant as established in 4.3.2.2, adjust the mount by means of the cant correction knob until the vertical reticle line of the sighting device is at least superimposed on or travels past one of the radial target lines. Readjust cant corrector to zero, then repeat operation in opposite direction. These operations will check for the minimum requirements of 3.4.2 and if the adjustment cannot be made it will be cause for rejection.

4.3.2.4 Elevation travel.- Adjust the cant corrector, elevation, and deflection knobs to zero position with the reticle lines of the sighting device superimposed on the vertical and horizontal lines of the target. Elevate the line of sight to maximum elevation by means of the elevation knob of the mount. Readjust the mount to zero elevation. Depress the line of sight to minimum elevation. The horizontal reticle line must at least superimpose on or travel past the 10 mil graduations to meet the minimum requirements of 3.4.3.

4.3.2.5 Deflection travel.- With the knobs of the mount positioned as

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in 4.3.2.4, traverse the mount in one direction to maximum deflection by means of the deflection knob of the mount, Readjust to zero deflection. Traverse mount in opposite direction to maximum deflection. The vertical reticle line must at least superimpose on or travel past the 10 mil graduations to meet the minimum requirements of 3.4.3.

4.3.2.6 Backlash.- Rotate the line of sight in a continuous clockwise direction by means of the deflection knob of the mount until the vertical reticle line is in coincidence with the vertical target line. Record reading on micrometer scale. Rotate line of sight past vertical target line (clockwise) a minimum of one full turn of the deflection knob, then by a continuous counterclockwise rotation of the deflection knob of the mount, bring the vertical reticle line in coincidence with the vertical target line and read micrometer scale. The difference between the recorded reading and second reading is the amount of backlash. Repeat this operation for elevation knob. The backlash shall not exceed the limits specified in 3.6.

NOTE: This test shall be conducted with the teeth in the knob assembly disengaged from the teeth on Cover C8226860. Rotation of the knobs must be uniform in one direction. The reticle line shall not override the target line during these operations. If override occurs this test is nullified and the mount must be reoriented and the operations specified in 4.3.2.6 will again be performed.

4.3.2.7 Locking levers.- Superimpose the reticle lines of the sighting device on the vertical and horizontal target lines. The locking levers shall be locked with the torque specified in 3.5. The results of this operation shall cause no change in elevation and azimuth alinement as indicated by the reticle lines of the sighting device on the vertical and horizontal target lines.

4.3.2.7.1 The elevation locking lever mechanism shall be locked the torque specified in 3.5 and the reticle lines of the sighting de superimposed on the vertical and horizontal target lines. The elev tion boresight knob shall be disengaged and the torque specified in 3.5 sha be applied to the elevation boresight knob. There shall be no movemen in elevation as indicated by the horizontal reticle line of the sighting device and the horizontal target line. Repeat these operations for the azimuth locking lever and the boresight knob. There shall be no movement in azimuth as indicated by the vertical reticle line of the sighting device and the vertical target line.

4.3.3 Operating knobs.

4.3.3.1 The elevation locking lever mechanism shall be locked by means of the locking lever, then unlocked. Torque shall be applied to the disengaged boresight knob. The torque required to rotate the boresight knob shall not exceed that specified in 3.7.1. Repeat this test for azimuth boresight knob.

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4.3.3.2 Torque shall be applied to the cant correction knob. The torque required to rotate the knob shall not exceed the limits specified in 3.7.2.

4.3.4 Environmental test.- The first ten (10) consecutive mounts and one (1) mount from each lot of twenty-five (25) produced thereafter that have passed all other requirements of this detailed specification shall be subjected to the requirements of 3.8.

4.3.4.1 Torque.- The mount shall be subjected to the temperatures specified in 3.8. While at the specified temperatures, torque shall be applied to the cant correction knob and each boresight knob (boresight knob in disengaged position). The torque required to rotate the knobs shall meet the requirements of 3.8. Torque test on boresight knobs shall be conducted with the locking lever mechanism unlocked.

4.3.5 Environmental failures.- Failure of any mount to meet the requirements of 3.8 shall require two (2) additional mounts to be selected from the same production lot and subjected to the test previously failed. Upon failure of either of these two additional mounts to meet the requirements, further acceptance of mounts will continue only by permission of the contracting officer.

4.4 Inspection of components and subassemblies.- Components and subassemblies shall be inspected in accordance with Table II. The acceptance and rejection numbers are expressed in terms of defective units. One or more defects on any unit shall constitute a defective unit. The sampling plans listed in Table II are not intended to preclude 100% inspection by the contractor if the contractor so desires.

TABLE II
Sampling Plan for Component Inspection

Lot Size (inclusive)	Sample Size	Accept Lot	Reject Lot
0-25	100% Inspection	0	1
26-35	20	0	1
36-50	25	0	1
51-150	30	0	1
151-over	30	1	2

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing, and marking shall be in accordance with Specification MIL-M-13093.

6. NOTES

6.1 Ordering data.- Invitation for bids and contracts shall specify the title, number and date of this specification, and type of packing.

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6.2 Definitions.

6.2.1 Gage.- A gage includes all precision standards and applicable accessories which are calibrated to one basic size to provide an accurate reference dimension.

6.2.2 Device, special testing (DST).- Special testing devices are designed to check function of an assembled unit.

6.2.3 Standard measuring equipment (SME).- Standard measuring equipment (SME) includes the common hand type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment.

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